

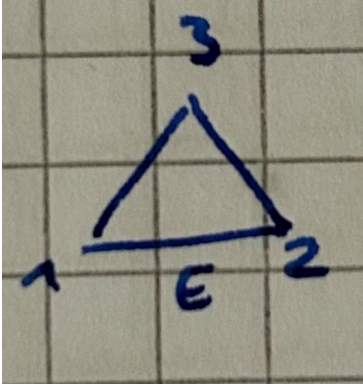
Calculations for the exact integration of $\int_E J_3 v_{\text{CR}}$

From the calculations in “integrateJ3vCR.pdf” we have the closed formula:

$$J_3 v_{\text{CR}} = a_1 \lambda_1 + a_2 \lambda_2 + a_3 \lambda_3 + 6(b_1 \lambda_2 \lambda_3 + b_2 \lambda_1 \lambda_3 + b_3 \lambda_1 \lambda_2) + \frac{27}{\sqrt{|T|}} \lambda_1 \lambda_2 \lambda_3 [\alpha(c_1 + c_2 + c_3) - \beta(c_1 \lambda_1 + c_2 \lambda_2 + c_3 \lambda_3)]$$

on a triangle. Assuming that the edge we want to integrate over the edge between nodes 1 and two as depicted below, the formula simplifies to:

$$J_3 v_{\text{CR}} = a_1 \lambda_1 + a_2 \lambda_2 + 6b_3 \lambda_1 \lambda_2$$



Using the general formula for barycentric coordinates:

$$\int_T \lambda_1^{\alpha_1} \lambda_2^{\alpha_2} = n! |T| \frac{\alpha_1! \alpha_2!}{(n + \alpha_1 + \alpha_2)!}$$

we get: $\int_E \lambda_i = \frac{|E|}{2}$ and $\int_E \lambda_1 \lambda_2 = \frac{|E|}{6}$. Therefore the result is:

$$\int_E J_3 v_{\text{CR}} = \frac{|E|}{2} (a_1 + a_2) + |E| b_3 = \frac{|E|}{2} (a_1 + a_2 + 2b_3)$$